Ephemeral Low-Rank Cores for Transformer FFNs: Representation Limits, Memory Accounting, and Controlled Evaluation

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Abstract

Input-conditioned low-rank modulation combines compact persistent factors with a generated diagonal core. Its representational, storage, and execution effects must be distinguished to establish an efficiency benefit. We study this separation for gated Transformer feed-forward networks (FFNs). We derive an output-subspace error floor that no controller can overcome while the final factor remains fixed, an activation-weighted constant-core control, and a matched-schedule bound on removable descriptor transfers. These results give falsifiable tests of when dynamic modulation and descriptor fusion are useful. We then reanalyze archived accelerator records and perform fresh software verification. In four recorded training-seed summaries for a Qwen2.5-1.5B FFN, the generated model improves relative reconstruction MSE only marginally at matched rank and is 1.95% worse at the closest reported parameter budget. Original execution reports and checkpoints for that family are unavailable, so these are auditable archive calculations rather than independently reproduced quality measurements. A same-run archived kernel summary reports 0.8234 versus 0.8509 ms for fused and materialized execution at 128 tokens, but persistent low rank takes 0.2719 ms. A later multicore variant fails its recorded repeated-call stability gate. Fresh CPU tests of the actual causal FFN reduce forward-retained nonparameter tensor storage by 13× under checkpointing, with identical outputs and gradients in nine small synthetic fixtures. The evidence supports precise mechanism and limitation statements; it does not yet establish a full-model quality–latency advantage.

1 Introduction

Transformer feed-forward networks contain large parameter matrices and several intermediate activations. Reducing one of these objects can change arithmetic, parameter capacity, memory transfers, and kernel launch overhead simultaneously. A speedup observed after a low-rank replacement therefore does not identify which change caused it. This distinction matters for conditional weight generation: a compact controller may add expressivity, while computing and materializing its outputs can cost more than the transfers that fusion removes.

We examine MWG-EW, a causal, token-conditioned diagonal modulation of persistent low-rank FFN factors. For a row vector x, one projection is

$$p(x) = \big((xU) \odot z(x)\big)V, \quad U \in \mathbb{R}^{d_{\text{in}} \times r}, \quad V \in \mathbb{R}^{r \times d_{\text{out}}}. \tag{1}$$

The algebra is closely related to feature modulation and hypernetworks. The systems question is whether the generated descriptor z should be stored as a global tensor or consumed immediately

where it is formed. We call the latter a *descriptor no-store contract*. The factors, controller parameters, and output activations retain their own lifetimes; the contract concerns only the named descriptor and a declared implementation boundary.

Three contributions organize this study. First, we derive representation and control results that separate changing rank from changing conditional coefficients: an output-subspace obstruction and an optimal constant-core projection control. Second, we give a memory-interface-specific accounting that separates parameter storage, generated cardinality, scheduled global transfers, and physical DRAM traffic. Third, we evaluate the resulting claims against a reproducible archive audit and new CPU software checks. Negative comparisons and failed validation gates are retained because they determine which deployment conclusions the evidence can support.

The most useful empirical conclusion is conditional. Generated modulation can approximately match a trained same-rank reconstruction baseline in the available summaries, but the strongest same-budget comparison favors the persistent baseline. Eliminating an explicit descriptor tensor is feasible at the source level; it does not imply that the complete FFN runs faster. Both conclusions are consistent with the analysis and motivate a smaller set of decisive experiments than a broad collection of disconnected speed and quality claims.

2 Relationship to Prior Work

Conditional parameters and feature modulation. HyperNetworks generate another network's parameters (Ha et al., 2017). FiLM applies conditional feature-wise affine transformations (Perez et al., 2018); Eq. (1) is multiplicative modulation in a rank-space representation. Learned static activation scales also appear in $(IA)^3$ (Liu et al., 2022). A static diagonal scale can be absorbed into a factor, whereas a token-conditioned scale generally cannot. LoRA (Hu et al., 2022) is a low-rank update to a retained base matrix, while the studied FFN replaces each dense projection by factors. These distinctions preclude treating the present algebra as an unprecedented form of generation.

The recent Ouroboros preprint uses a hidden-state controller to modulate SVD-initialized low-rank bases within recursive Transformers (Jaber and Jaber, 2026). It is especially close algebraic prior art. Our scope is the lifetime and controlled assessment of FFN descriptors, including the limits imposed by a fixed output subspace. We make no claim of priority over input-conditioned diagonal modulation.

I/O-aware execution and recomputation. FlashAttention analyzes and reduces transfers between memory levels (Dao et al., 2022). FlashMHF combines dynamically weighted FFN subnetworks with online SRAM computation (Zhang et al., 2025); it is a relevant future systems and model-quality comparator. Activation checkpointing trades retained tensors for recomputation (Chen et al., 2016), and Checkmate optimizes rematerialization schedules (Jain et al., 2020). Our CPU checkpoint experiment applies this established technique to the actual MWG module. It verifies compatibility and instrumentation rather than proposing a new checkpointing algorithm.

Compression and fair comparison. Quantization methods such as GPTQ and AWQ reduce the representation cost of weights (Frantar et al., 2023; Lin et al., 2024). They remain important deployment baselines. Rank reduction, quantization, modulation, and fusion should be compared at matched task quality and reported budgets. The archived experiments here do not support numerical superiority over those methods. The output rank bound uses the classical Eckart–Young–Mirsky approximation result (Eckart and Young, 1936; Mirsky, 1960).

Table 1: Which mechanism a controlled contrast can identify.

Question	Matched contrast	Required invariant
Does rank reduction help?	Dense versus trained persistent factors	Task metric, data and optimization budget
Does conditioning help?	Dynamic versus constant-core/static control	Parameter or compute budget; causal evaluation
Does descriptor fusion help?	Same dynamic model, materialized versus fused	Checkpoint, inputs, output tolerance and timing boundary
Does recomputation help?	Eager versus checkpointed same module	Parameters, input, loss and all gradients

3 Model and Execution Contract

3.1 Causal gated FFN

Let d be the model width and m the FFN intermediate width. The dense reference, using row-vector conventions, is

$$f(x) = \big[\mathrm{SiLU}(xW_g) \odot (xW_u)\big]W_d, \quad W_g, W_u \in \mathbb{R}^{d\times m}, \quad W_d \in \mathbb{R}^{m\times d}. \tag{2}$$

In the factorized model, each W_p is replaced by Eq. (1). For token state $x_{b,t}$ the controller produces all three rank vectors:

$$h_{b,t} = \mathrm{SiLU}(x_{b,t}A + a_0), \tag{3}$$

$$(z_u, z_g, z_d)_{b,t} = \mathbf{1} + \alpha \tanh(h_{b,t}C + c_0), \tag{4}$$

$$\widetilde{f}(x) = p_d\big(\mathrm{SiLU}(p_g(x)) \odot p_u(x); z_d(x)\big). \tag{5}$$

Here $A \in \mathbb{R}^{d\times h}$ and $C \in \mathbb{R}^{h\times 3r}$; all factors and controller parameters are trainable persistent state. The down-projection core is generated from the original pre-FFN state x, while its projection input is the gated intermediate. The controller has

$$P_G = (d+1)h + (h+1)3r \tag{6}$$

parameters, including biases. For equal fp32 factor/controller precision, total parameter bytes are $4[3r(d+m) + P_G]$. Optimizer states and saved activations are separate quantities.

The implemented causal mode uses a distinct core for each token ($q = 1$). It reads no future hidden states and pools no companion examples. If a future implementation shares a core across $q > 1$ tokens, both group boundaries and its context must be determined from information available by the first token's pre-FFN state. Pooling the complete group's teacher-forced states would violate this condition. A batch of B examples of length S has

$$n = B\lceil S/q \rceil, \qquad 1 \le q \le S, \tag{7}$$

groups under batch-isolated grouping. Cached decoding has $S = q = 1$ and $n = B$; we do not label a 128-token call as single-token autoregressive decoding.

3.2 The actual fusion boundary

The custom Ascend operator receives rank activations $a = xU$, controller hidden states h, and the controller head's parameters. It forms the head affine result, applies the bounded transform in

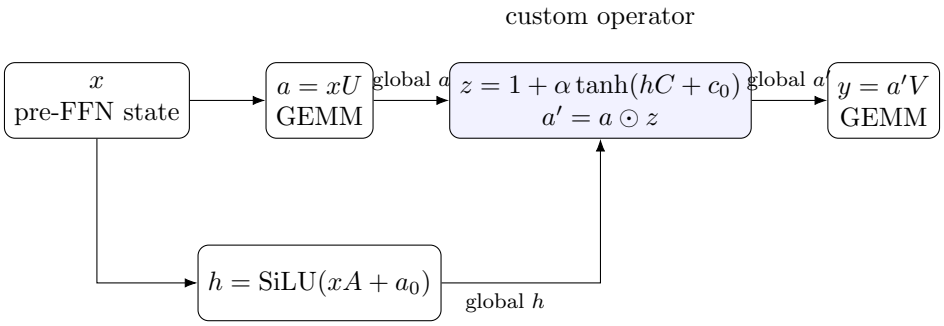


Figure 1: Implemented boundary for one projection. Only the descriptor z is consumed without an explicit global descriptor tensor. Source-level absence of a descriptor store does not prove absence of compiler or runtime spills.

Eq. (4), and multiplies a by the resulting scale. The descriptor is not returned as an explicit global tensor. The operator *does* write $a \odot z$ to global memory, and the U and V GEMMs run outside it. Consequently, neither the entire projection nor the entire FFN is scratch-resident.

We distinguish three verification levels. A source audit checks that z is not returned, aliased, or explicitly stored. A compiled-kernel inspection checks workspace and spill behavior for the exact binary. A device-counter comparison measures transfers under a declared workload and cache state. The first does not imply the other two. Binary hashes bind inspected code to an artifact; they do not prove that a reported remote run actually used it.

During training, autograd may retain controller intermediates and descriptors. The inference no-store contract therefore does not automatically extend to backward. Deterministic rematerialization is one possible training strategy; Section 7.4 tests software checkpointing, not the custom kernel's backward implementation.

4 Representation Limits and a Stronger Control

4.1 Fixed output-subspace obstruction

A conditional core changes coefficients in fixed rank directions. In one projection, the effective matrices belong to the linear span of the r outer products formed by columns of U and rows of V; the span has dimension at most r. The input-output mapping may still be nonlinear because z depends on x. The final output restriction is stronger and independent of the controller's complexity.

Theorem 1 (Output-subspace obstruction)**.** *Let $V_d \in \mathbb{R}^{r \times d}$ be the fixed final factor in Eq. (5), let $\mathcal{S}$ be its row space, and let $P_{\mathcal{S}}$ be the orthogonal projector onto $\mathcal{S}$. Stack N target FFN outputs into $Y \in \mathbb{R}^{N \times d}$ and corresponding model outputs into $\widehat{Y}$. Then, for any controller,*

$$\|Y - \widehat{Y}\|_F^2 = \|Y(I - P_{\mathcal{S}})\|_F^2 + \|YP_{\mathcal{S}} - \widehat{Y}\|_F^2 \geq \|Y(I - P_{\mathcal{S}})\|_F^2. \tag{8}$$

If the final factor is optimized subject to rank at most r, every candidate still satisfies

$$\|Y - \widehat{Y}\|_F^2 \geq \sum_{j>r} \sigma_j(Y)^2. \tag{9}$$

Proof. Every output row is a coefficient row multiplied by V_d, so $\widehat{Y}P_{\mathcal{S}} = \widehat{Y}$. The two terms in the decomposition $Y - \widehat{Y} = Y(I - P_{\mathcal{S}}) + (YP_{\mathcal{S}} - \widehat{Y})$ have orthogonal row components, giving the

Pythagorean identity. Also $\operatorname{rank}(\widehat{Y}) \le r$; the best rank-$r$ approximation of Y has squared Frobenius error equal to its singular-value tail (Eckart and Young, 1936; Mirsky, 1960). □

The theorem is a structural consequence of classical low-rank approximation, not a new general approximation theorem. It explains a concrete limitation: context-dependent coefficients cannot recover target components orthogonal to the final persistent factor. The lower bound is evaluated on *uncentered* target outputs. For residual-block outputs, subtract the unchanged skip connection before applying it. An arbitrary coefficient matrix can attain the best projection onto a fixed subspace, but the restricted FFN/controller need not attain it. The theorem does not say that controllers are useless within that subspace, nor that all layers of a residual Transformer share one subspace.

A useful future diagnostic reports three quantities on fixed evaluation activations: actual reconstruction error, the fixed-factor floor in Eq. (8), and the best rank-r output floor in Eq. (9). Their gaps distinguish coefficient approximation, subspace selection, and unavoidable rank loss. These real-model floors cannot be recovered from scalar MSE summaries alone and are not reported as measured here.

4.2 An activation-weighted constant-core baseline

A shuffled context control changes the input to the controller and may change the optimization process. It does not by itself show whether a learned dynamic core can be replaced by an efficient constant. A sharper projection-level control fits a constant using training/calibration activations only.

Proposition 1 (Optimal frozen-projection constant)**.** *For fixed row-vector projection inputs $a = xU$, fixed V, and frozen dynamic outputs $z(x)$, write $z(x)$ and c as column vectors in this proposition. Define $Q(x) = \operatorname{diag}(a)VV^\top \operatorname{diag}(a)$, $H = \mathbb{E}[Q(x)]$, and $b = \mathbb{E}[Q(x)z(x)]$. Assume the needed second moments are finite. The constants minimizing*

$$R(c) = \mathbb{E}\big[\|(a \odot (z(x) - c)^\top)V\|_2^2\big] \tag{10}$$

solve $Hc = b$. A minimum-norm solution is $c^* = H^\dagger b$; all other solutions add a vector in $\ker(H)$.

Proof. Q is positive semidefinite and $R(c) = \mathbb{E}[z^\top Qz] - 2c^\top b + c^\top Hc$. If $v \in \ker(H)$ then $\mathbb{E}[v^\top Qv] = 0$, so $Qv = 0$ almost surely and $v^\top b = 0$. Thus b belongs to the range of H, the normal equations are consistent, and completing the quadratic gives the stated minimizers. □

The empirical version replaces expectations by calibration averages; its constant folds into $U \operatorname{diag}(c^*)$. An ordinary mean of z ignores the activation and output metric Q. This is a diagnostic application of weighted least squares, with no claim of novelty for the normal equations. Its optimum is calibration-local, with no automatic held-out guarantee. In the three-stage FFN, replacing earlier cores changes later inputs, so separately fitted projection controls require a new full-FFN and full-model evaluation. No constant-core quality improvement is inferred from the present archive.

4.3 Sensitivity and invariances

Let $A = \phi(XW_g)$, $B = XW_u$, and tildes denote approximations fixed within one group. If ϕ is L_ϕ-Lipschitz, $\|W_p - \widetilde{W}_p\|_F \le \delta_p$, $\|B\|_{\max} \le M_u$, and $\|\widetilde{A}\|_{\max} \le M_g$, then

$$\|Y - \widetilde{Y}\|_F \le \|W_d\|_2 \|X\|_2 (M_u L_\phi \delta_g + M_g \delta_u) + \|\widetilde{A} \odot \widetilde{B}\|_2 \delta_d. \tag{11}$$

To verify it, use $C - \widetilde{C} = (A - \widetilde{A}) \odot B + \widetilde{A} \odot (B - \widetilde{B})$ and $Y - \widetilde{Y} = (C - \widetilde{C})W_d + \widetilde{C}(W_d - \widetilde{W_d})$, then apply submultiplicativity. Token-conditioned matrices permit pointwise bounds followed by aggregation; a fixed-matrix SVD optimum alone does not establish conditional learnability.

Two paired rank scales are redundant since $s \odot s'$ is a single scale. Removing them does not make the complete factorization identifiable. For invertible diagonal D, $U \mapsto UD$ and $V \mapsto D^{-1}V$ preserve Eq. (1) because diagonal matrices commute. All storage comparisons count actual parameters rather than counting equivalent parameterizations as distinct capabilities.

5 Memory Accounting at a Declared Interface

5.1 Four quantities that must remain separate

Dense parameter cardinality is $P_D = 3dm$, and persistent factor cardinality is $P_F = 3r(d + m)$. A call generates $D_z = 3nr$ diagonal scalars. These are counts of objects; adding them does not produce a lower bound on DRAM traffic. For example, a warm cache can suppress DRAM reads of persistent factors, while multiple tiles can load the same factor repeatedly.

Let b_w, b_G, b_z be bytes per factor, controller parameter, and descriptor scalar. Parameter storage is $b_w P_F + b_G P_G$. Let L_{common} denote transfers across one declared global-memory interface that are shared by two implementations. It includes persistent-factor loads, controller loads, source activations, scaled rank activations, and outputs according to the actual schedule. It is not interchangeable with parameter storage. Physical DRAM bytes require device counters and depend on caching and writeback.

Proposition 2 (Removable final-descriptor transfers)**.** *Suppose two schedules are identical except that one writes and reloads each of the $3nr$ final descriptor scalars exactly once while the other consumes those values locally. With $C_z = 3nrb_z$,*

$$L_{\text{mat}} = L_{\text{common}} + 2C_z, \qquad L_{\text{fused}} = L_{\text{common}}. \tag{12}$$

If the common schedule includes one full cold read of every factor, $L_{\text{common}} \geq 3r(d+m)b_w > 0$, and therefore

$$\frac{L_{\text{mat}}}{L_{\text{fused}}} \leq 1 + \frac{2nb_z}{(d+m)b_w}. \tag{13}$$

Proof. The only differing transfers are a store and load of C_z bytes each. Divide $L_{\text{common}} + 2C_z$ by L_{common} and substitute its lower bound. □

This is a scheduled-byte statement, not a physical-HBM or latency bound. Different tiling, cache reuse, launch counts, controller-logit temporaries, casts, or arithmetic violate the matched-schedule premise. In particular, an eager materialized implementation may store more intermediates than the final z; a fused implementation can remove those too. The bound isolates only the final-descriptor round trip. It prevents attributing all improvements from an unrelated multi-operation rewrite to one missing tensor.

The corrected regime-map implementation reports scratch cardinality separately. Its partial no-store global ledger is $e_F b_w P_F + e_G b_G P_G + c_x BSd\, b_x$; materialization adds $2C_z$. Common output and other activation transfers are not included, so these partial totals must not be presented as whole-FFN traffic. The implemented ledger uses $b_z = b_w$; Eq. (13) allows distinct widths. Version 2 explicitly changes the previous mixed-interface totals.

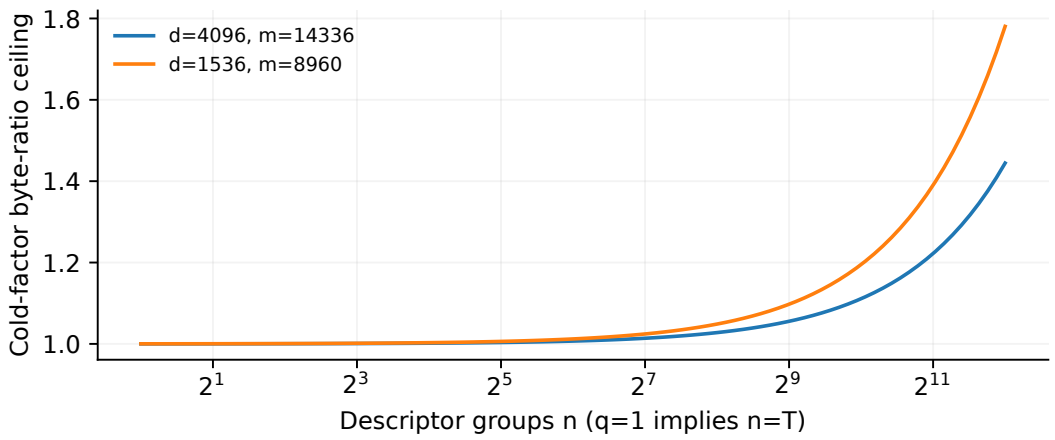


Figure 2: Analytic upper ratio in Eq. (13), assuming equal factor/descriptor widths and one cold full-factor read. This is not measured speedup. For $d + m = 18432$, the ceilings are 1.01389 at $n = 128$ and 1.44444 at $n = 4096$. Rank cancels in this isolated model.

5.2 Full-factor generation and reuse

Generating all factor pairs per group emits $3nr(d + m)$ scalars instead of $3nr$. Relative to the dense parameter cardinality, the one-copy ratio is $n\gamma$, where $\gamma = r(d + m)/(dm)$. For positive integer dimensions and $0 < \gamma < 1$, let $n_{\max} = \lceil 1/\gamma \rceil - 1$ and $g_{\max} = \lfloor n_{\max}/B \rfloor$. Strict cardinality reduction is possible exactly when $g_{\max} \geq 1$, and then it requires

$$q \geq \lceil S/g_{\max} \rceil . \tag{14}$$

Indeed, $B\lceil S/q \rceil \gamma < 1$ is equivalent to $\lceil S/q \rceil \leq g_{\max}$. At $(d, m, r, B, S) = (4096, 14336, 128, 1, 1024)$ this gives $q \geq 43$; at $(1536, 8960, 384, 1, 1024)$ it gives $q \geq 342$. These are optimistic generation-volume thresholds, not traffic or quality claims. The causal grouping restriction still applies.

5.3 Training and routing

Under dense synchronous data parallelism, communicating gradients for fewer persistent parameters can reduce communication volume. The generated values are functions of controller parameters and are not independent optimizer parameters. This explains a low-rank communication saving regardless of whether descriptors are materialized during inference.

A runtime router saves work only when it dispatches the selected branch before that branch's weights and kernels are consumed. Computing both branches and then selecting with `torch.where` provides output selection without the claimed branch-skipping savings. A quality threshold selected on evaluation data also gives an optimistic policy estimate. A future router evaluation must use separate training, calibration, and final evaluation data, fixed routing rules, and whole-call transfer measurements. This paper makes no population tail-risk guarantee or operational router claim.

6 Evidence Protocol and Provenance

The empirical questions are: (Q1) does conditioning improve reconstruction at a matched budget; (Q2) does fusion improve the same conditional computation; (Q3) can software rematerialization

Table 2: Evidence classes after the October 22, 2026 audit.

Evidence	Available artifacts	Supported scope
Qwen FFN quality	Four per-seed summaries and duplicate aggregate record	Arithmetic and within-archive contrasts; original run provenance incomplete
Four-arm v1 timing	One same-run summary per geometry	Recorded medians; no local per-iteration uncertainty
Multicore v4	Timing/equivalence records plus failed stability record	Exploratory timings; reliable speedup claim blocked
DDP optimizer timing	Three complete JSON run reports	Short kernel-execution timings; numerical health unknown
CPU checkpointing	New source, nine cases, hashes and tests	Actual-module saved-storage and gradient verification
Algebra checks	New script and 50 seeded fixtures	Numerical consistency of identities; no model-quality evidence

preserve the computation and gradients; and (Q4) do quality and efficiency hold simultaneously in a complete model? Table 2 makes the available evidence explicit.

Archive integrity. The canonical quality family is `asi2_rank512_control_20260922T080042Z`. Its two local files have SHA-256 hashes recorded in the audit. Their per-seed records agree exactly. Hash matching establishes integrity of those files, not independent proof of execution. The canonical directory lacks original per-seed execution reports, trained checkpoints, and complete immutable data manifests. Several alternative closing-run report fragments cannot be parsed as complete JSON and are excluded. The revision preserves historical artifacts and records these limitations.

Quality protocol reported by the archive. The target is the layer-16 FFN of Qwen2.5-1.5B-Instruct, with $d = 1536, m = 8960$, fp32, ranks 384, 401, and 512, 5000 steps, learning rate 2×10^{-4}, batch size one, and 128 evaluation batches. Training seeds are 2026–2029. Factor seed 4242 and activation seed 1729 are shared. Consequently these are four training-seed records from shared data preparation, not four independently disjoint datasets. The record names separate train/test text files and a near-duplicate filter threshold 0.82; the available archive does not establish the exact realized splits or checkpoint identity.

The evaluator reports relative MSE as total squared reconstruction error divided by target squared magnitude for equal-sized evaluation batches. In notation, $\mathrm{RelMSE} = \|\widehat{Y} - Y\|_F^2 / \|Y\|_F^2$ for a nonzero target. It measures FFN output reconstruction, not language-model perplexity or task accuracy. All within-rank contrasts use matching seed labels from one family. The shown intervals are $\overline{\Delta} \pm t_{.975,3} s_\Delta / \sqrt{4}$, with $t_{.975,3} = 3.1824463$. They are unadjusted descriptive intervals under an independent normal paired-run model, whose adequacy cannot be established with four archived summaries. No preregistered equivalence margin, population claim, or multiple-comparison significance claim is made.

Table 3: Archived layer reconstruction: mean ± sample standard deviation across four training-seed records. Lower MSE is better. Storage includes factors and, when applicable, the controller under the stated $h = 192$ assumption. All rows are recomputed summaries, not newly executed training.

Model	Rank	fp32 MiB	Relative MSE
Static SVD	384	46.125	0.836904 ± 0.001721
Trained persistent	384	46.125	0.449632 ± 0.001417
Shuffled controller	384	48.099	0.449086 ± 0.001302
Token controller	384	48.099	0.448878 ± 0.001323
Static SVD	401	48.167	0.826043 ± 0.001690
Trained persistent	401	48.167	0.440310 ± 0.001374
Shuffled controller	401	50.178	0.439704 ± 0.001242
Token controller	401	50.178	0.439508 ± 0.001273
Static SVD	512	61.500	0.754122 ± 0.001742
Trained persistent	512	61.500	0.383298 ± 0.001197
Shuffled controller	512	63.757	0.381959 ± 0.001184
Token controller	512	63.757	0.381983 ± 0.001191

Table 4: Paired generated-minus-control MSE contrasts. Negative values favor the generated arm. Intervals are descriptive and unadjusted; four records do not certify equivalence, generalization, or independence.

Rank	Control	Mean difference	Descriptive 95% interval
384	Trained persistent	-0.000754	$[-0.001363, -0.000146]$
384	Shuffled controller	-0.000208	$[-0.000261, -0.000154]$
401	Trained persistent	-0.000802	$[-0.001450, -0.000155]$
401	Shuffled controller	-0.000196	$[-0.000257, -0.000134]$
512	Trained persistent	-0.001315	$[-0.001378, -0.001252]$
512	Shuffled controller	$+0.000025$	$[-0.000007, +0.000056]$

7 Controlled Results

7.1 Rank, capacity, and conditioning

Table 3 recomputes every value from the same four seed records. Parameter storage is reconstructed using Eq. (6) and the code-default hidden width $h = 192$; this assumption is explicit because the canonical record omits the resolved width. It is not a measurement of peak memory. A fixed additive 2 MiB controller approximation is inaccurate across ranks and is not used.

At rank 384 the generated-minus-persistent difference is -0.0007545; its descriptive interval excludes zero (Table 4). This numerical finding does not establish a meaningful task-quality improvement. The matched-rank generated-minus-shuffled differences are approximately -0.000208 and -0.000196 at ranks 384 and 401, about 0.05% of the control MSE. At rank 512 the difference is approximately $+0.000025$ with an interval spanning zero. Shuffling is therefore consistent with little observed dependence on the correct token context in these summaries. It does not prove that the learned controller is constant, or predict its behavior on other distributions.

The closest parameter-budget comparison has generated rank 384 at 48.099 MiB and persistent rank 401 at 48.167 MiB. Their mean relative MSE values are 0.448878 and 0.440310: the generated arm has a 1.95% relative penalty. These are near-matched rather than exactly equal budgets. Increasing persistent rank to 512 further reduces MSE, but that comparison consumes a larger

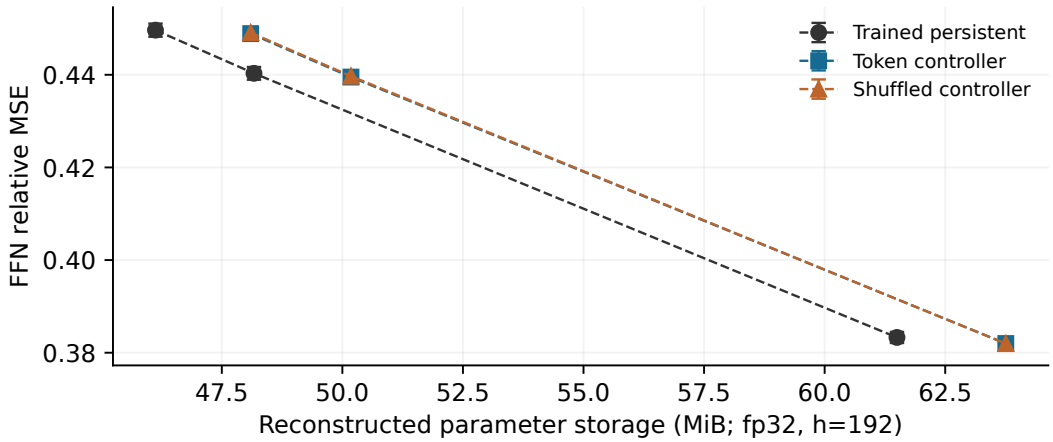


Figure 3: Recomputed archive MSE versus reconstructed parameter storage. Error bars are across-seed sample standard deviations, not corpus-level confidence intervals. Lines join observed rank settings only and do not imply measured interpolation. Persistent low rank is a competitive capacity control.

Table 5: Same-run archived v1 timings in milliseconds. The final column is fused latency divided by materialized latency. The probe includes host-side caching of tiling/workspace and the fp16 controller-weight image.

Tokens	Dense	Persistent	Materialized	Fused v1	Fused/materialized
128	0.4611	0.2719	0.8509	0.8234	0.96768
4096	–	–	1.3932	2.6821	1.92514

budget and cannot isolate conditioning. Figure 3 keeps that capacity cost visible. The observed pattern motivates the constant-core control and output-subspace diagnostics; neither has yet been measured on these trained checkpoints.

7.2 Fusion versus materialization

Table 5 uses only same-run pairs in the shipped v1 summary. The probe has $(d, m, r, h) = (4096, 14336, 128, 128)$, $B = 1, S = 128$, fp16, $\alpha = 0.05$, and seed 1234 on Ascend 910B2. The described protocol uses 30 warmup launches, 200 timed iterations, and five repeats with synchronized host timing. The dense matrices in this harness are exact products of the shared factors; this is an execution control, not an independently trained full-rank checkpoint. The dynamic arms use the same factors and controller, but their outputs need not match the unmodulated persistent arm. Those original full reports are designated remote in the summary; local per-iteration traces are unavailable. Thus the table reports recorded medians without newly asserted statistical uncertainty or cache-residency claims.

At the probe, fusion reduces recorded latency by 3.23% relative to materialization, while it remains 3.03× slower than persistent low rank. At $T = 4096$ it is 1.93× slower than materialization. Both comparisons matter: the first isolates an implementation difference in a generated model; the second asks whether that model is efficient against the practical static baseline. No measured quality-matched deployment improvement follows.

The v4 multicore record contains exploratory v1/v4 timing ratios of 1.13, 1.38, 1.70, 2.14, and

Table 6: Archived optimizer-inclusive synchronized execution, mean ± sample standard deviation across three reports. Ratios use means. These microbenchmarks lack recorded finite-loss/gradient/update health gates, so they support execution-time accounting only.

Arm	Time (ms)	Dense/arm ratio
dense	24.5906 ± 0.5983	1.000
mwg r32	6.0953 ± 0.2943	4.034
mwg r64	6.1529 ± 0.4865	3.997
mwg r96	6.5256 ± 0.9379	3.768
mwg r128	6.0112 ± 0.0996	4.091
mwg r192	5.9397 ± 0.1202	4.140
mwg r256	5.9841 ± 0.2141	4.109
mwg r384	5.9887 ± 0.2008	4.106
mwg r512	7.2919 ± 1.2630	3.372

3.03 for $T = 128, 512, 1024, 2048, 4096$. However, its 1000-call stability test ends with `pass=false` and a nonfinite final error. The cause is unresolved; it cannot be assigned to an environment transient from this record alone. The audit therefore marks the stability gate **FAIL** and excludes v4 from validated efficiency claims. There is also no matched v4-versus-materialized complete-FFN experiment. Multiplying ratios from different runs does not close this gap.

7.3 Optimizer-inclusive execution timings

Table 6 recomputes three archived four-device run reports at $d = 4096, m = 14336, B = 2, S = 512$, fp16. The timed region includes zeroing gradients, forward, squared loss, backward, DDP synchronization, and AdamW update. There are three warmup and five timed training iterations per repeat. The reported statistic is the maximum across devices of each device's mean time, then summarized across three repeats. It is not a sample of independent end-to-end training runs to convergence.

The dense mean is 24.5906 ms, and low-rank ratios range from 3.372 to 4.140. The benchmark changes parameter count and controller behavior and does not run the inference descriptor no-store kernel. These ratios cannot identify a lifetime effect. Moreover, the FP16 optimizer benchmark records no finite-loss, finite-gradient, or finite-parameter gate. The revision therefore does not interpret those timings as healthy training, convergence acceleration, or a quality-preserving replacement of an 8B-parameter language model. The phrase "8B-scale geometry" refers only to matrix dimensions.

7.4 Fresh CPU rematerialization verification

To provide complete local evidence for one narrower claim, we executed the actual `LowRankFFN` with `token_diag_core`, comparing ordinary autograd with non-reentrant PyTorch checkpointing. Fixtures use synthetic random inputs and randomly initialized factors, $d = 32, m = 64, r = 8, h = 16$, $B = 2, S = 16$, fp32, and $\alpha = 0.2$. Residual stacks $x \mapsto x + \widetilde{f}(x)$ have depths 1, 2, and 4; seeds are 2026–2028. Each arm begins from an identical deep copy. Conditioning response and controller gradients are required to be nonzero.

Saved-tensor hooks count unique underlying nonparameter storages retained by the FFN forward. Aliases of parameters are excluded; shared storage is counted once. Caller-owned inputs and full storage underlying saved views are included. Loss-function saved tensors are outside the measured region. This quantity is not allocator peak memory, total training memory, or hardware traffic.

Table 7: New CPU software measurements, three seeds per depth. Exact byte counts were identical across these seeds. Output, loss, input-gradient and all parameter-gradient maximum absolute differences were zero in all nine cases. The controller was active and all checked tensors were finite.

Depth	Eager saved bytes	Checkpoint saved bytes	Ratio	Cases
1	53,248	4,096	13.0	3
2	106,496	8,192	13.0	3
4	212,992	16,384	13.0	3

Python 3.12.14 and PyTorch 2.14.1+cpu were used with one CPU thread. The raw JSON records exact source hashes, initial-state hashes, per-case tensor checks, and the measurement boundary. Tests reject missing or nonfinite gradients and verify parameter-alias exclusion. Table 7 supports software rematerialization for these fixtures. It supplies no evidence about NPU checkpoint speed, peak memory, or trained-model quality. The previously quoted 24.5–71.8-fold memory headline is not carried forward because the available legacy evidence does not establish an instrumented actual-MWG measurement.

7.5 Algebra checks

Fifty seeded float64 NumPy fixtures verify the output-subspace identity and compare the constant-core normal equations with an independently constructed least-squares system. Maximum rowspace residual is 1.42×10^{-14}; maximum relative orthogonal-decomposition residual is 8.88×10^{-16}; constant-core coefficients agree within 3.67×10^{-15}. The fitted constant has no greater calibration loss than the unweighted mean core in these fixtures. These are checks of the derivations, explicitly separated from real-model experiments and from the proofs themselves.

8 Limitations and Decisive Follow-Up Experiments

The theoretical analysis is general within its stated architecture and scheduling assumptions. The available quality evidence is narrow and has incomplete execution provenance. The accelerator evidence comes from one device family, with small repeat counts and different configurations for quality and systems probes. The new CPU experiment establishes a local software property. These pieces do not establish Q4, a simultaneous full-model quality–latency benefit.

A decisive next study should freeze a trained causal checkpoint, its tokenizer, dataset splits, and evaluation harness; then measure both quality and runtime for that same artifact. The primary quality comparison should include trained persistent low rank at exact or explicitly near-matched parameter and compute budgets, a calibrated constant-core control, and relevant quantized baselines. Held-out perplexity and task metrics must accompany FFN reconstruction. Rank floors require stored evaluation activations and target outputs; a controller claim should explain the reducible error above those floors.

For execution, the generated/materialized and generated/fused arms must share checkpoint, input, precision, warmup, cache policy, and complete-FFN timing boundaries. Alternate paired launches, report timing distributions, and verify against an independent high-precision reference before and after extended execution. Gate all timing claims on finite outputs, gradients where relevant, healthy repeated calls, and supported shapes/devices. Record device counters and distinguish logical transfers from DRAM measurements. Test true cached decode ($S = 1$), prefill, varying batch sizes, and the serving stack if a serving claim is made. The current failed v4 stability check must be

resolved before that variant can enter an efficiency comparison.

Finally, choose quality margins, primary workloads, repeat counts, and stopping rules before collecting the next comparison. A successful result requires both acceptable held-out quality and an observed efficiency gain on the same implementation. If constant-core or persistent-factor controls dominate, the correct conclusion is a limitation of dynamic modulation in that regime. The mechanism and negative evidence remain useful, but they should not be renamed as an established deployment advantage.

9 Conclusion

Ephemeral diagonal cores expose a narrow and testable execution choice within a familiar conditional low-rank model. Their fixed final factor imposes an output-subspace floor, and eliminating a descriptor tensor removes only the transfers associated with that tensor under a matched schedule. The archive shows modest same-rank reconstruction differences, a near-budget disadvantage, and no validated complete-model efficiency gain. New actual-module CPU checks establish a reproducible rematerialization property with exact gradient agreement. These results identify what is supported and which joint experiment would determine whether the approach is useful in deployment.

Reproducibility and artifact availability

The canonical manuscript, audit script, algebra checks, CPU lifecycle test, and archived source records are in the accompanying research directory. `scripts/build_journal.sh` rebuilds the audit-derived tables, figures, and author-review PDFs. The audit JSON contains exact input hashes and an explicit false submission-readiness flag for the unresolved empirical gates. `paper/REVIEW_20261022.md` maps each review finding to its correction or remaining requirement. Data and checkpoint availability limitations are stated in the evidence protocol, rather than treating a file hash as execution proof.

AI assistance disclosure

Generative AI tools assisted technical review, mathematical derivations, implementation, verification, analysis, and manuscript revision. Newly executed synthetic fixtures are labeled as such; archived accelerator records were reanalyzed without claiming new device execution. The author is responsible for reviewing the arguments, artifacts, and disclosures before any submission.

Broader impact

The intended application is more efficient neural-network execution. Energy savings, production cost reductions, and improved model capabilities have not been measured here. Incomplete numerical checks or overstated performance claims could mislead deployment decisions, which motivates explicit evidence boundaries and preservation of failed gates.

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